



How cells move

A new discovery indicates that the force cells apply onto the surface under it determines cell shape and motion. <https://digit.in/feb20-15>



Detailed molecular maps captured

Using a process called MSI, scientists captured intricate details of the molecular makeup of animal tissue. <https://digit.in/feb20-16>

initial two papers reported 31,000 and 26,588 protein-coding genes. A more complete draft of the genome appeared in 2004 which estimated that a completed catalogue would contain 24,000 protein-coding genes. The Ensemble human gene catalogue described in that paper had 22,287 protein-coding genes and 34,214 transcripts.

THE EXPANDING NUMBER OF RNA GENES AND THE INVENTION OF RNA-SEQ

The year 2008 saw the invention of the RNA-seq which was designed to improve the quantification of gene expression, also further improving the ability to detect transcribed sequences, both coding and non coding. Many of the noncoding transcripts contained introns and were quite long, thereby leading them to be called lincRNAs (long intervening non-coding RNAs) which was later shortened to lncRNAs thereby dropping the “intervening”. lncRNA databases have grown drastically over the last decade which led to current human gene catalogues containing more RNA genes rather than protein-coding genes.

EXPANDING NUMBER OF SPLICE VARIANTS

RNA-seq revealed alternative splicing, alternative transcription initiation, and alternative transcription termination was far more frequent than prior knowledge, possibly affecting as many as 95 percent of human genes. These findings imply that, despite knowing where all the genes are, there still remains considerable work to discover all the isoforms of those genes, which leads to more work to determine whether these isoforms have any function or if they just represent splicing errors.

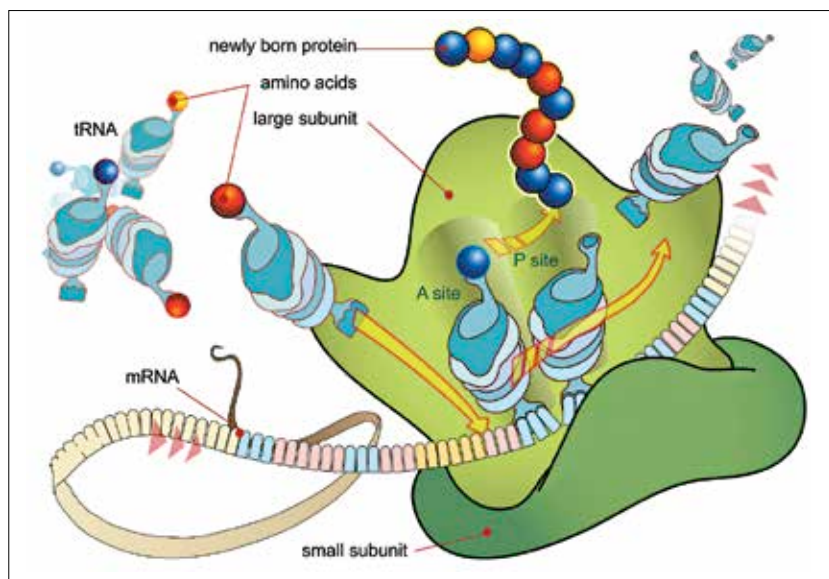
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We still have to face the challenge of identifying all human genes. For the past 15 years, the gene lists has been predominantly controlled by just two groups : RefSeq, maintained by the National Center for Biotechnology Information (NCBI) at NIH, and Ensem-

bl/Gencode, maintained by European Molecular Biology Laboratory (EMBL). With passing time and not much progress, the two catalogues have had hundreds of disagreements between their lists of protein-coding genes, thousands of inconsistencies between their lncRNAs, and multiple categories of genes where they end up separating even further, resulting in them not even agreeing on the type of gene. The catalogues are still evolving, resulting in hundreds of protein-coding genes being added or being deleted from the Gencode list. These disagreements

the dilemma for the users in case of the database to be referred. Despite the fact that the large number of genes may include more false positives, the larger set of genes will still prove to be useful to the studies of human diseases that have still not yet found a genetic cause. The CHESSE gene set version 2.0 is not yet in its final state and will see improvement in the years to come.

Long story short, it is still inconclusive as to how many genes we have in our body, although progress is still being made. Many genes, lncRNAs in particular, appear to be highly tissue-



RNA Translation

contribute to the ongoing challenge of creating a much more comprehensive human gene catalogue.

Given the abundance of disagreements in the existing databases, in the year 2017, a new human gene database called CHESSE was created. CHESSE used a massive RNA-seq collection to compile once again all of the transcripts from a broad survey of human tissues. The CHESSE gene set is intended to provide a far more comprehensive collection of human genes by adding more than 10,000 new gene isoforms and a lesser number of new genes to the existing databases. CHESSE, by design, includes all of the protein-coding genes from both Gencode and RefSeq so as to prevent

specific. Surveying all human cell types thoroughly might take many more years. We cannot be sure whether all human genes and transcripts have been discovered. Our knowledge regarding the gene catalogue of other animals are plants is far less, although our knowledge is rapidly improving at a decent pace. Our knowledge of the Human genes is far more richer than it was during the onset of the HGP. Hence, the inability to find an answer to this simple question proposed by the Human Genome Project does not translate to its failure. With the advancements in technology and medicine over the past decade, we can say optimistically that one day, this number will be eventually narrowed down. ■